

Payment.java

Teacher's Guide & Speaker Notes for Presentation

TEACHING THE CLASS: The Abstract Blueprint

```
public abstract class Payment
```

How to explain this: When you show this file on screen, start by explaining the word "abstract". Use an analogy: You can't just build a generic "Building", you have to build a "House" or an "Office". Similarly, we never process a generic "Payment", we only process specific things like Credit Cards. This class is just a blueprint that holds the common data they all share.

TEACHING THE VARIABLES: Encapsulation & Static

```
private static int ticketCounter = 1000;  
protected double amount;
```

How to explain this: Point out the word "static" first. Explain to your audience that "static" means this variable is shared by the entire application. It's not tied to one payment. That is how we can increment it safely to generate unique Transaction IDs (TXN1000, TXN1001) without duplicates.

Next, point out "protected". Tell them this is Encapsulation in action! We hide the amount from the outside world so a hacker or a bug can't change the price, but we let the child classes (like PayPal) access it.

TEACHING THE CONSTRUCTOR: Automation

```
public Payment(double amount, String currency)
```

How to explain this: Explain that whenever a new payment is born, this block of code runs automatically. It acts as a gatekeeper (throwing an error if the amount is negative) and acts as an automator (recording the exact date and time immediately).

TEACHING THE METHOD: The Contract

```
public abstract boolean process(BiConsumer<String, String> log);
```

How to explain this: This is the most important line in the file! Ask the audience to notice that there is NO code inside this method. Explain that by marking it "abstract", we are creating a strict rule: "Any payment type created in the future MUST have a process() method." This guarantee is what makes the rest of our system work seamlessly.